Lab 07 — Create Reusable Financial Ratio Functions

Python Programming for Finance · TGS-2026065049 · LO4

Scenario

A bank reporting team calculates margin, ROE and ROA in several reports and needs one tested implementation.

Goal

A ratios module returning named results with validation and tests.

Tools

Python, dataclasses, unittest

Step-by-step

1. Review the required function contract and units.

2. Implement safe_divide() with a defensible zero-denominator result.

3. Implement net_margin(), roe() and roa().

4. Return a dictionary of named ratios from calculate_ratios().

```
python solution.py
```

5. Test typical, zero and negative cases.

```
python -m unittest test_solution.py
```

6. Document assumptions in evidence.txt.

Acceptance criteria

The sample produces margin about 55.95%, ROE about 18.09% and ROA about 1.53%; tests pass.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.